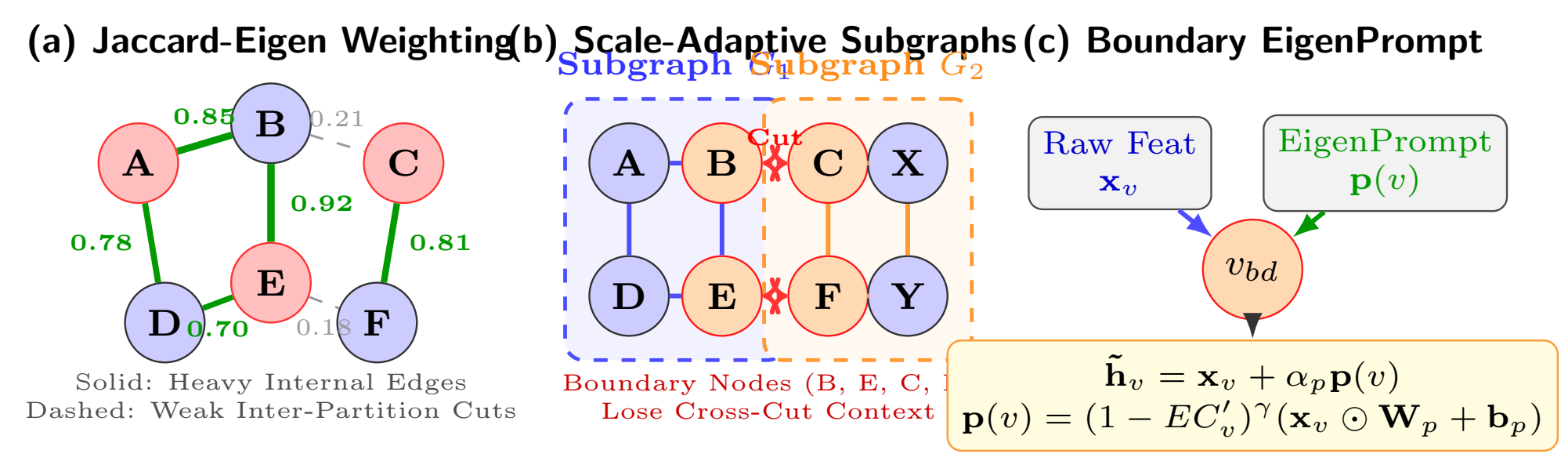

(a) Jaccard-Eigen Weighting
(b) Scale-Adaptive Subgraphs
(c) Boundary EigenPrompt
0.85
0.21
0.92
0.78
0.81
0.70
0.18
A
B
C
D
E
F
Solid: Heavy Internal Edges
Dashed: Weak Inter-Partition Cuts
Subgraph G_1
Subgraph G_2
Cut
X
Y
Boundary Nodes (B, E, C,
Lose Cross-Cut Context
Raw Feat
$\mathbf{x}_v$
EigenPrompt
$\mathbf{p}(v)$
v_{bd}
$\tilde{\mathbf{h}}_v = \mathbf{x}_v + \alpha_p \mathbf{p}(v)$
$\mathbf{p}(v) = (1 - EC'_v)^{\gamma}(\mathbf{x}_v \odot \mathbf{W}_p + \mathbf{b}_p)$